Mobile Communications Market Development

October, 2025

Ⅰ. Data Pre-Processing

Data Pre-Processing in Power BI Was Faster and Easier Than in Python

Filtered out columns, added region and income groups using composite keys → Pivoted, merged tables and created new indicators and measures → Unpivoted each indicator into a new table, grouped by totals and averages → Appended all into one table → Filtered out data for correlations in copied tables

- Data was obtained in *.zip from the public website of **International Telecommunications Union**, https://datahub.itu.int/ except for summary table with country classification - https://datahub.itu.int/dashboards/idi/?e=ISR&y=2025
- Zip files are saved and unpacked locally because downloading them from the website is inefficient and slower
- Data includes **3 dataframes** – mobile voice indicators, population and subscribers and country income and regional classification
- The dataframes were merged, data for 2003-2008 was deleted due to data unavailability for some indicators, countries with more that one zero except the last year were dropped. The remaining data for 137 countries in incomplete but is quite representative for the objective
- New indicators were calculated further in ITU_Utilities – 1. **Market Size** (ARPU * Subscribers *12) and 2. **Penetration Rate** (Subscribers / Population)
- ITU_Utilities also calculates **means for country aggregates classified by income, region and World** ARPU and Penetration Rates and also includes the **totals** for the other country-aggregate indicators
- The resulting master file for further manipulations was appended to the initial population file
- The analysis here is **based only on mobile voice data**, for shortcut the missing conclusions are provided by the author without references intentionally, the focus of this is python-based data processing capacity rather than full-scale financial analysis

II. Data Analysis

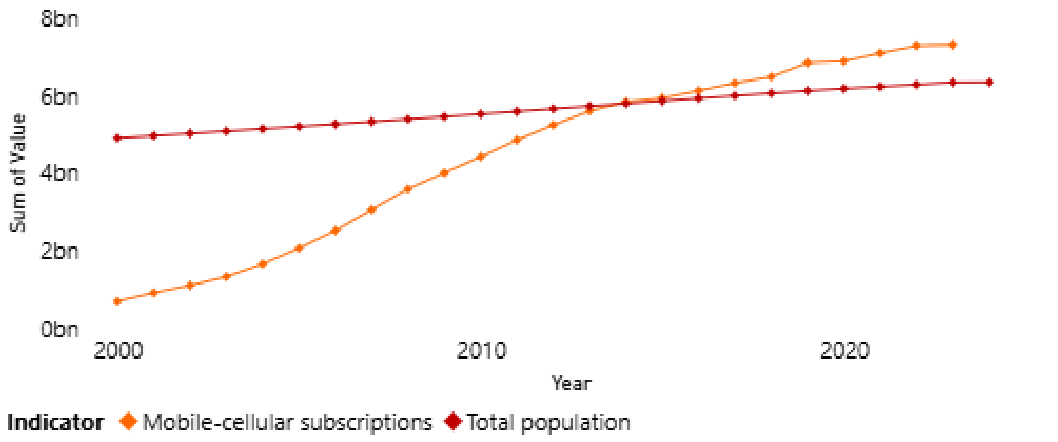
Global Mobile Voice Data Market Overview

- This presentation is focused on mobile communications market to demonstrate Power BI application skills; marketing or investment conclusions, while correct, are not the purpose of this presentation
- The golden era of mobile voice market was disrupted by messengers and social media around 2011-2013
- The growth of subscribers in the voice segment led to the decline of mobile voice market due to the decline of ARPU
- The global mobile penetration rate exceeded 100% in 2015 with significant differences among the rich and poor countries
- Telecoms and TMT sector revenues overall continued to grow but with the shift from voice to data and content segments
- Part of revenues was lost to free of charge and “secure” messengers (WhatsApp), social media and mobile apps driving EBITDA margins from the peaks of about 52%-58% (higher than FAANNG – Facebook, Amazon, Apple, Netflix, Nvidia, Google) down to 32%-38% range and even lower
- Mobile operators are seeking for new content- and service- based growth models to avoid becoming a low-margin data traffic pipeline with varying but limited success (from negative to less than 10% EBITDA margin effect)

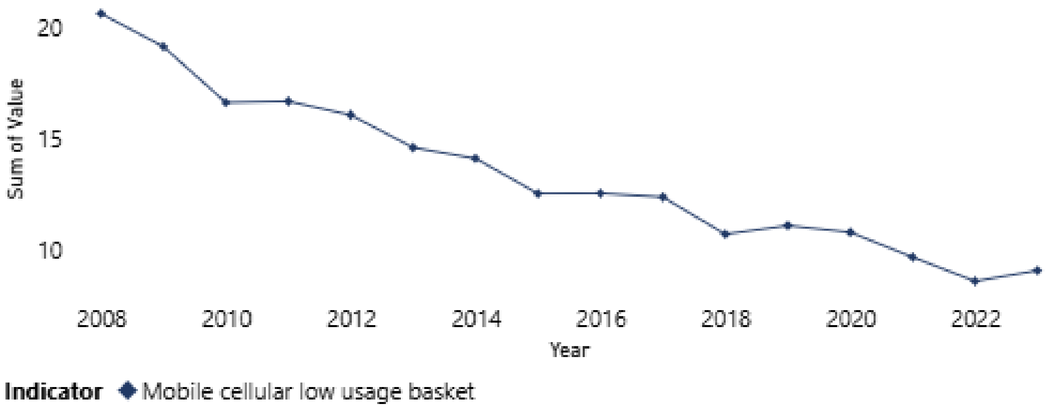
Mobile Telecommunications Market Overview

- Mobile voice data market was the fastest growing industry in the world ever
- By 2014 the global penetration rate exceeded the global population due to several numbers for voice data. Other factors, not related to voice services, also include wireless data modems, data only services and Internet of things (IOT) with individual additional SIM cards
- Average Global ARPU enabled the market growth driven both by the economies scale, technological improvements, competition, pay-as-you-go tariffs, transition from voice to lower cost data traffic in voice over LTE/Internet (VOLTE/VOIP) in mobile networks and data in messengers
- Those changes were different for individual countries and regions

Global Population and Subscribers Development



Global Mobile Voice Average Revenue per User (ARPU) Development



Source: ITU

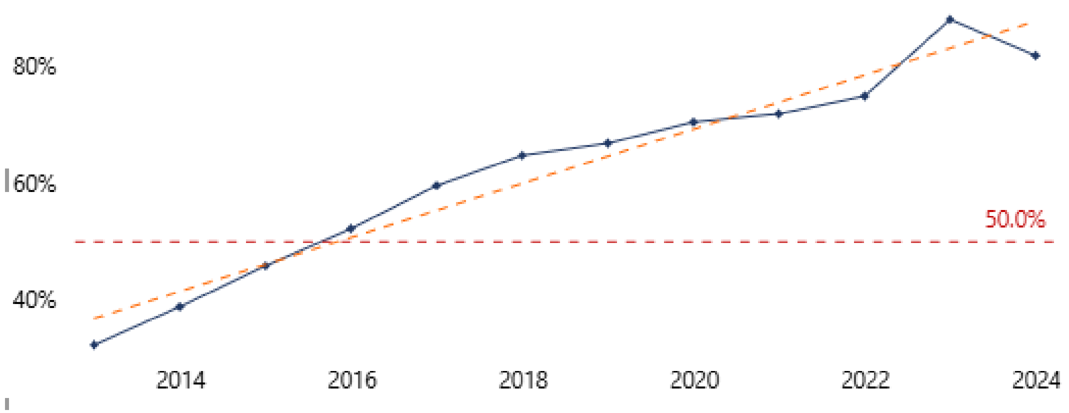
Cannibalization of Voice Calls by Mobile Internet

The tipping point from voice-only to mobile internet occurred in 2016 when the number of mobile internet users exceeded 50%

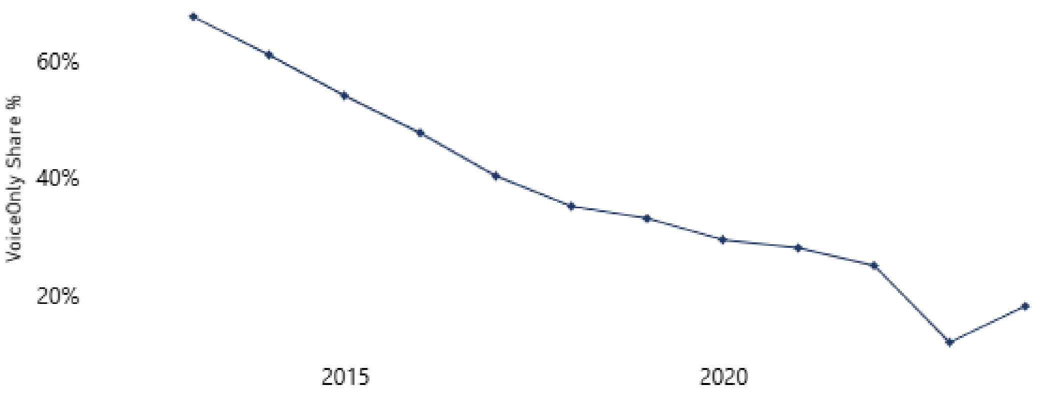
$$\textbf{Share\_MBB} = \frac{\text{Active MBB subs (per 100)}}{\text{Mobile-cellular subs (per 100)}} \in [0, 1]$$

- Mobile internet usage was driven by net affordability and reliability
- Mobile broadband penetration rate increased from ~23% to ~81% by +12% per year on average
- High income level countries passed 50% level of using mobile internet in 2013
- Mid-income countries passed the 50% level in 2017 and lower-income countries passed the mark in 2023
- The share of voice only reduced from 69% in 2013 to 18% in 2024 with the annual decline of ~10% on average

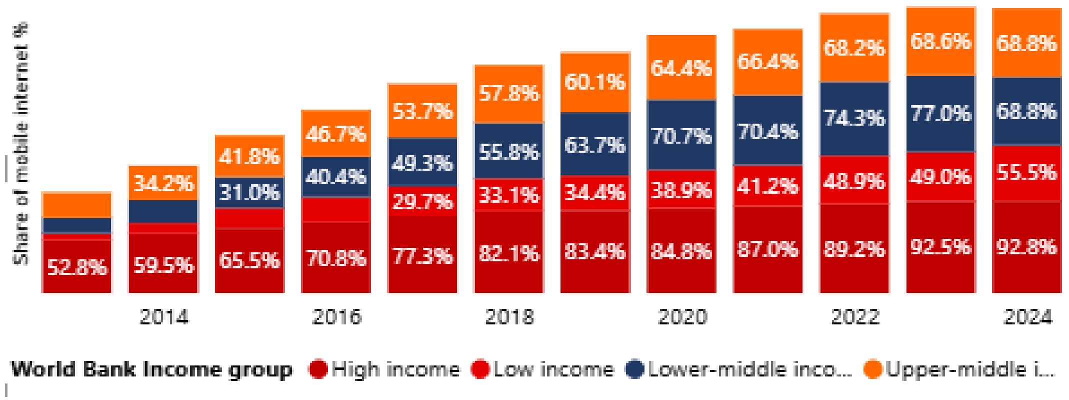
Growth of Mobile Internet Penetration



The Decline of Voice-Only Share of Services



Mobile Internet Penetration by Country Income Groups



Source: ITU

Price and Availability Metrics

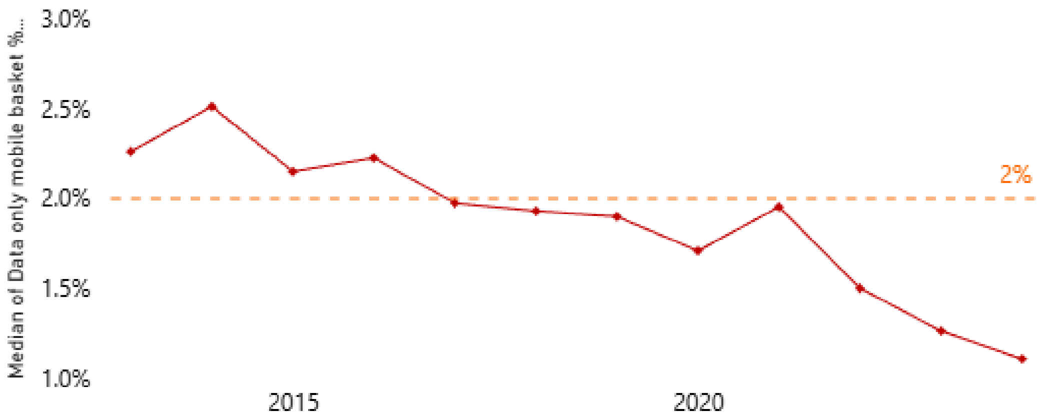
Data Affordability Increased as the cost of packaged mobile subscription fell from 2.26% in 2014 to 1.1% in 2024

- Affordability index shows the declining share of data basket in the real average personal monthly income

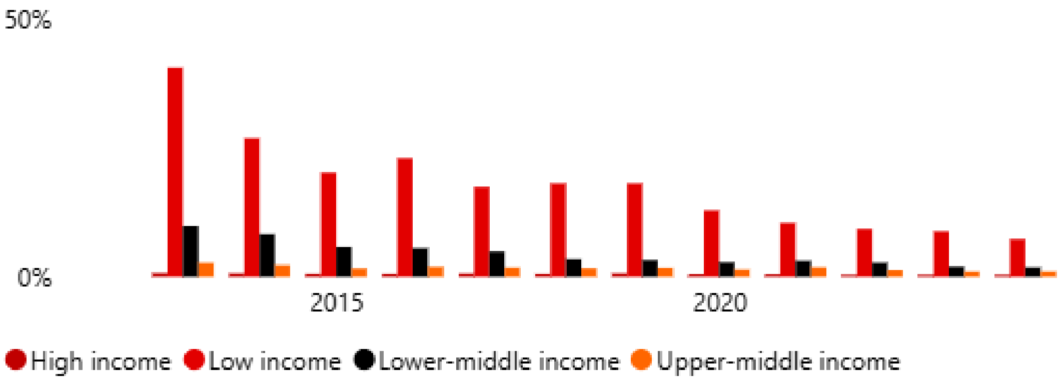
$$\mathbf{Aff\_Data\_\%GNI} = \frac{\text{Price of data basket per month}}{\text{GNI per capita}/12} \times 100$$

- The share of a basic mobile data plan in monthly per capita income in 2014-2024 fell from 2.5% to 1.1% making mobile internet significantly more affordable
- The 2% dotted line is a frequently used "target" affordability
- The share of mobile data-only internet in personal income for low income countries in 2014-2024 dropped from 10% to 7% and for lowest income countries from about 40% to 20%
- For low income countries while the affordability has improved the accessibility remains an issue
- Data consumption grew exponentially driven by smartphones, 4G/5G, video content, tariff bundles

Cost of Mobile Data-Only Basket as % of GNI



Cost of Mobile Data-Only Mobile Basket as % of GNI



Source: ITU

Substitution of Voice and Fixed-Line by Mobile Internet

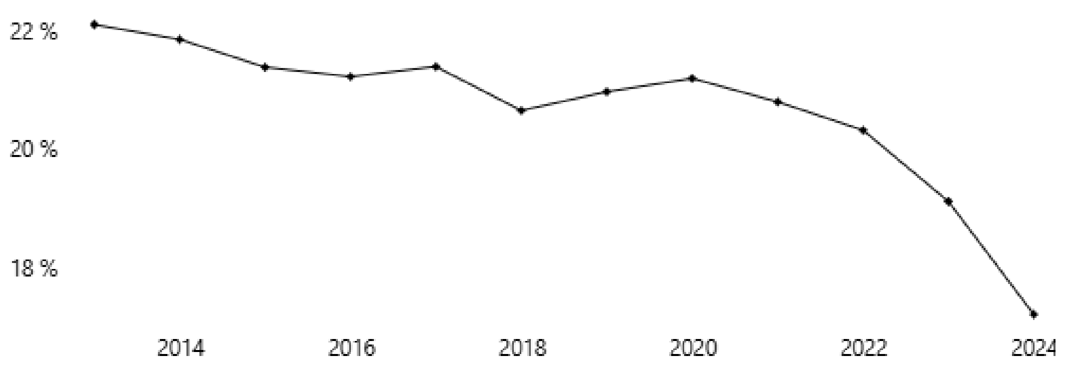
Data and mobile broadband have become more available

- Relative Price Index (Data vs Voice calls)

$$\mathbf{RPI} = \frac{\text{Data-only MBB basket (PPP\$)}}{\text{Mobile cellular low-usage basket (PPP\$)}}$$

- The Relative Price Index (Data/Voice) shows that mobile data prices are significantly lower than voice services and keep declining because the routing of data packages is more cost efficient than the commutation of voice channels
- Absolute prices in PPP$ are gradually declining, improving internet affordability

Data / Voice Relative Price Index

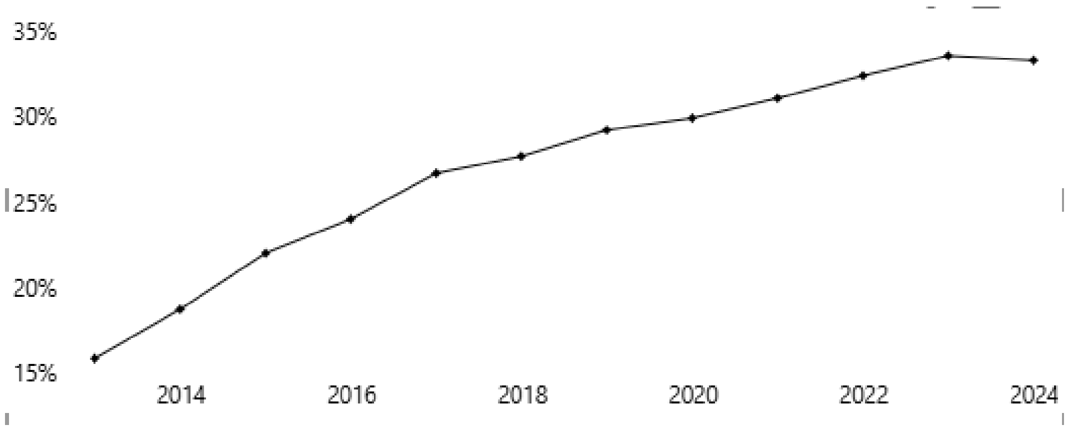


- Fixed-Mobile Substitution

$$\mathbf{MobileShare\_BB} = \frac{\text{Active MBB per 100}}{\text{Active MBB per 100} + \text{Fixed-BB per 100}}$$

- Data and mobile broadband have become more available with the share of mobile broadband reaching ~30% of all broadband connections
- MobileShare_BB = share of mobile broadband subscriptions among all broadb and subscriptions (mobile + fixed), per 100 inhabitants
- MobileShare_BB 30% ⇒ approximately third of broadband connections in the country are mobile
- Customers are increasingly using 4G/5G mobile Internet smartphones, inexpensive packages

Mobile Broadband Growth



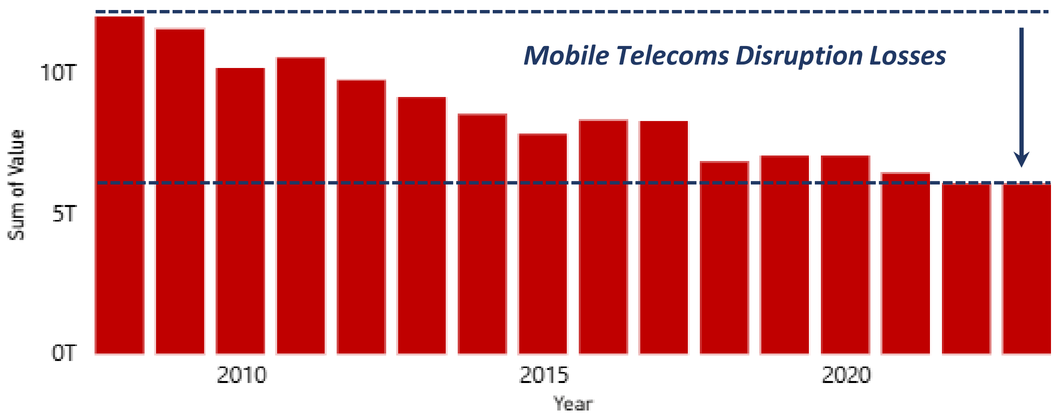
Source: ITU

Redistribution of Revenues in the TMT Sector

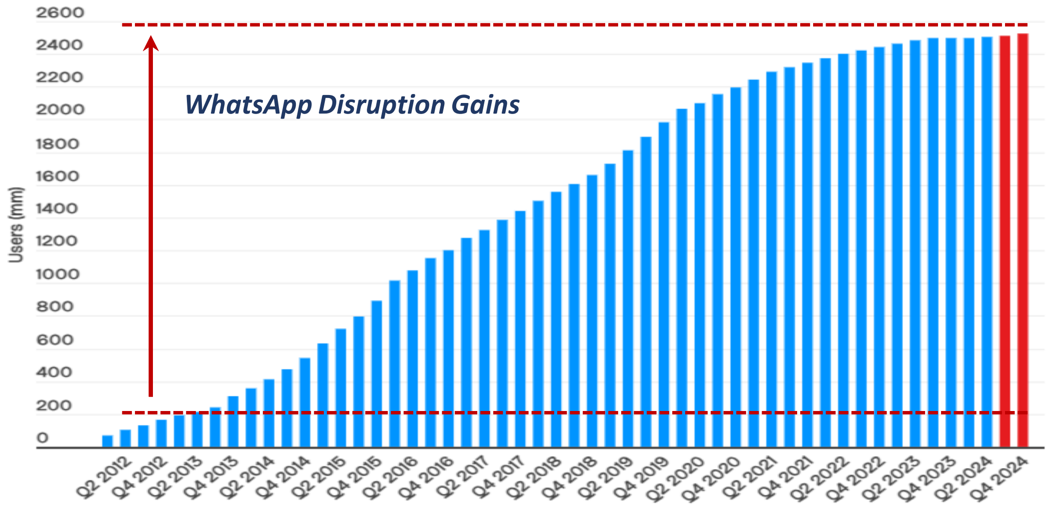
TMT Sector continued showing strong growth with structural changes - messengers and social networks disrupted the telecom sector

- Just like mobile telecoms disrupted the fixed-line operators in 2000s, messengers and social networks (Facebook Group) disrupted the mobile operators
- Since 2010s due to the competition flat rate package tariffs have prevailed in mobile telecoms affecting both ARPU and cost
- Value added and ARPU was disrupted by WhatsApp free voice calls, while Facebook and Google have dominated in the ad generating traffic segment
- In 2020s AI has disrupted the social media … yet TMT sector remains the top performer over the last 40 years

Mobile Telecoms Voice Market Development, 2008 - 2024

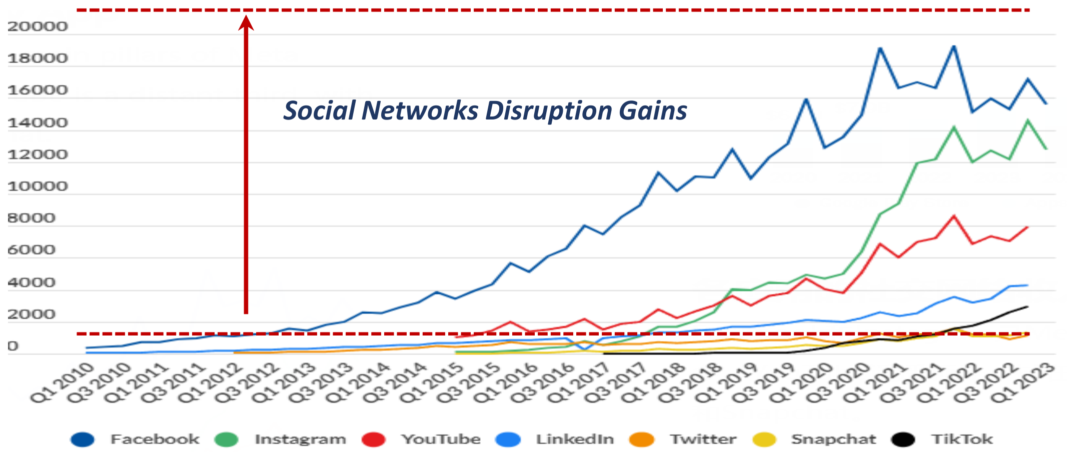


WhatsApp Users 2012 – 2024



[WhatsApp Revenue and Usage Statistics (2025) - Business of Apps]

Social Media Stats Worldwide, 2009 – 2023



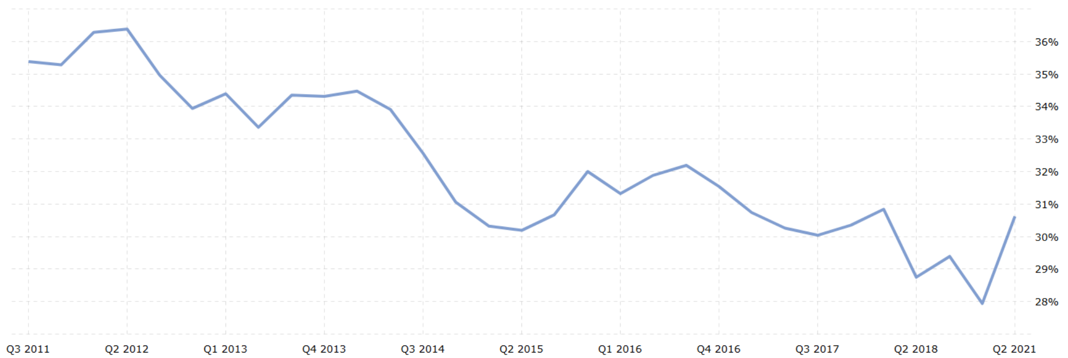
Source: ITU, Business of Apps, FoxData

[Fostering High Growth for Social Media Applications: Trends and Strategies-Appranking]

EBITDA Margin Effect in Developed and Emerging Countries

The disruption drove mobile operators’ EBITDA margins about 20%-25% down on average, varies between developed and emerging markets

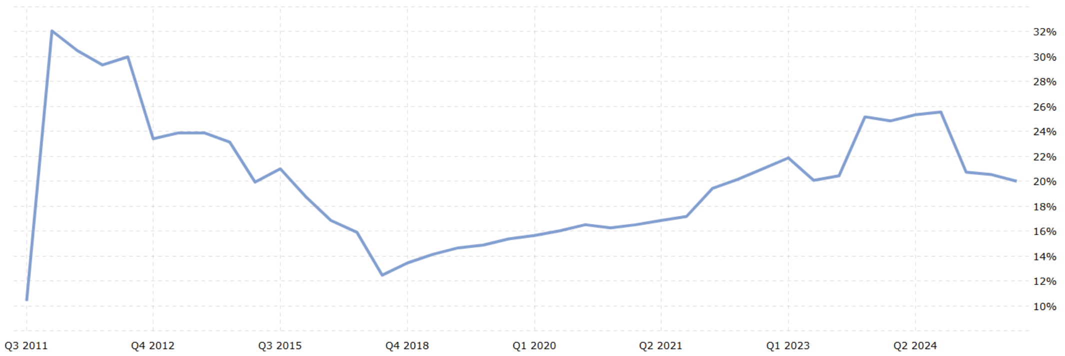
NTT DOCOMO, Japan, EBITDA Margin Development, 2011 - 2023



Telefonica Spain EBITDA Margin Development, 2011 - 2025

Verizon EBITDA Margin Development, 2011 - 2023

America Movil EBITDA Margin Development, 2011 - 2023



Source: www.macrotrends.net

Key Findings & Recommendations

KEY FINDINGS

- The global mobile voice services peaked in 2010-2013
- Mobile operators were disrupted by no fee communication apps, WhatsApp (messengers), and media content, Facebook (social media), around 2012
- Overall the TMT market continued to grow with revenues re-distributed between telecoms and media

RECOMMENDATIONS

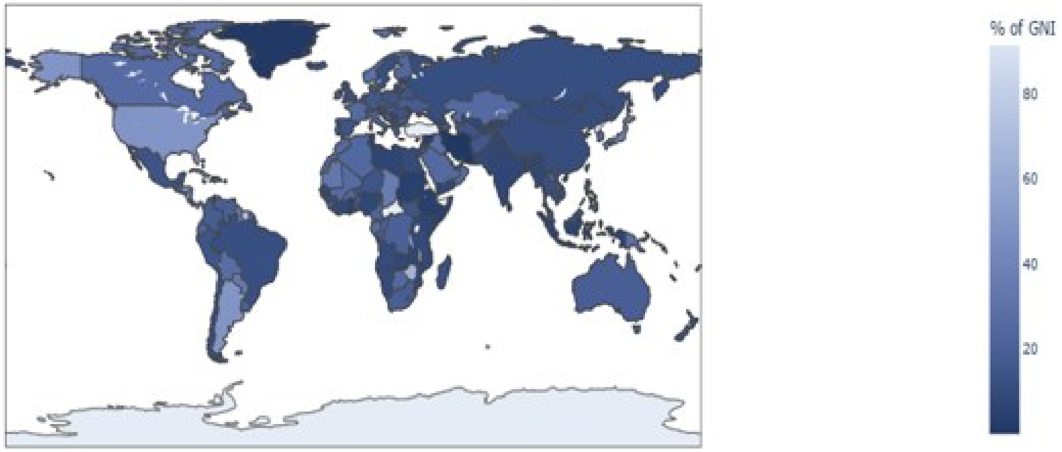
- Offer targeted tariff packages and personalized services to increase user revenues (ARPU)
- Develop higher-margin content and value-added service ecosystems to increase revenues and margins
- Careful implementation of AI tools to cut costs, create content, improve customer service and security

Investment Potential

Despite the market maturity mobile communications market provides strong investment and development potential in some emerging markets

- By 2023 the cost of mobile and internet access has become acceptable globally
- The remaining digital divide still provides opportunities for strategic investment
- Funding from development organizations, private investments and government support improve infrastructure, expand coverage, and accelerate the digitalization
- Increasing access to mobile communications provides opportunities for residents to study, work, business, and receipt of services
- Communications provide multiplication effect for economic growth for both individuals and countries
- Market capitalization growth potential still provides strong investment potential in some less developed parts of the world, at least lowest 20 countries by penetration

Cost of Mobile Access (% GNI), 2023



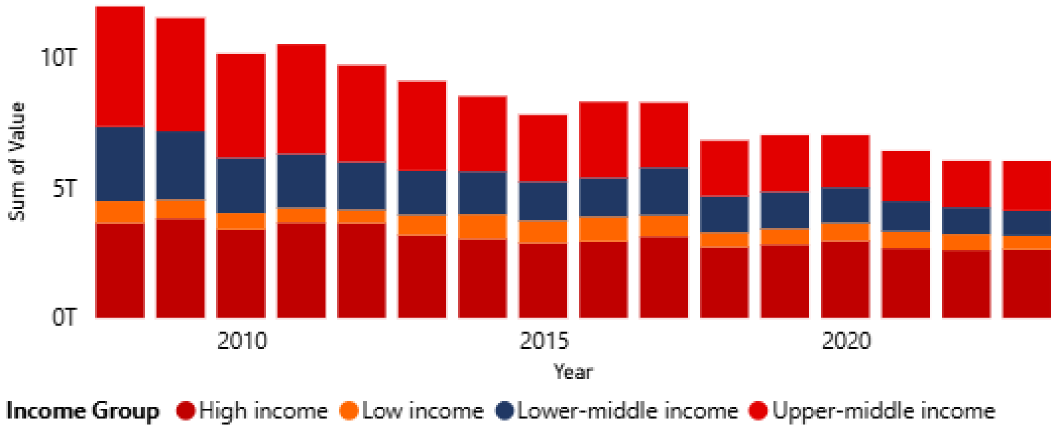
Lowest 20 Mobile Penetration Countries, 2023

Source: ITU

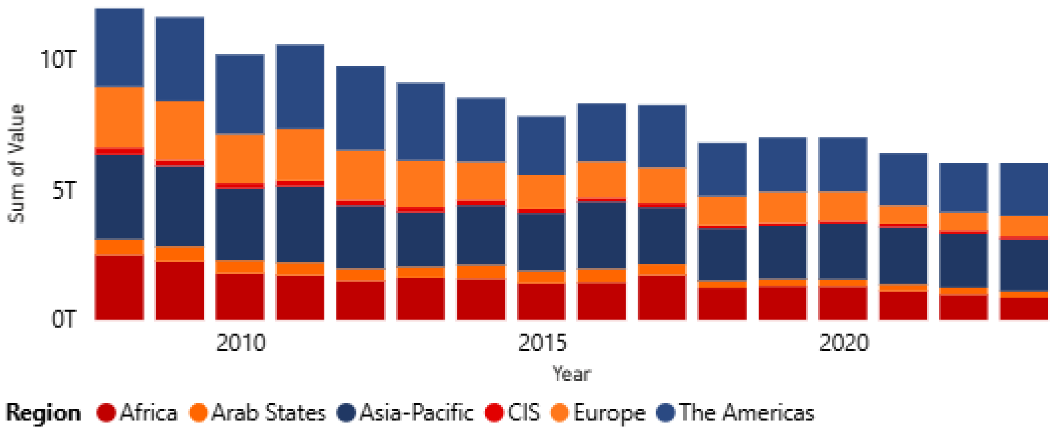
Mobile Voice Market Size Overview

- Consistent with the global slide regional charts reflect a strong decline in mobile voice market in 2010s
- The decline was stronger in the higher income economies due to the shift to data and messengers
- Asia was the only region, where mobile voice market still continued growing, which is partly explained by network coverage, catch up growth but also cultural preferences
- Since 2012 mobile operators were actively changing their business models to retain revenues and margins with very different success, the overall EBITDA margins erosion was roughly 15%-20% on average (not the focus of this presentation)

Mobile Voice Market Size in Key Economies by Income Split



Mobile Voice Market Size in Key Economies by Regional Split

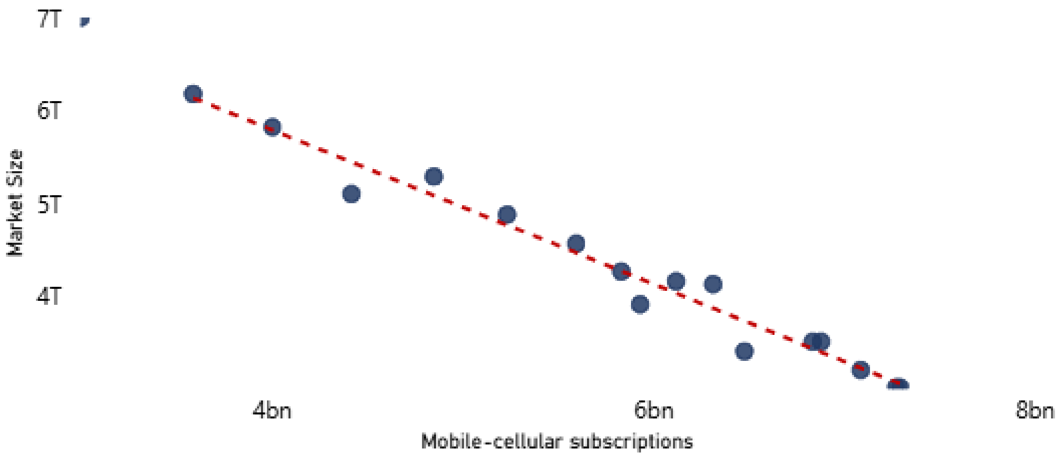


Source: ITU

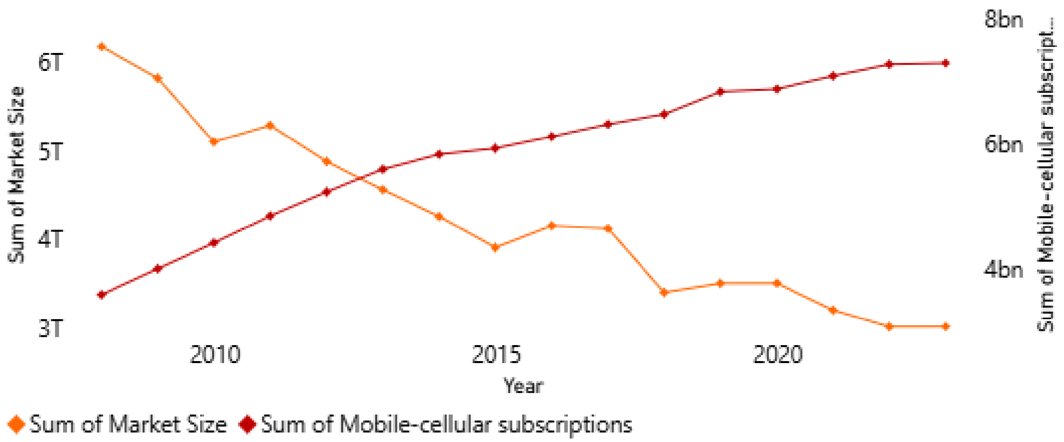
Growth Drivers: Subscribers

- The growth in the number of subscribers was the key driver for the global mobile voice market since the start of the market in 1990s
- However, the correlation for 2008 – 2023 was negative (0.75) with the key reasons being:
  - Most of the market growth happened in late 1990s -2008, which we don't see in the available data
  - The mobile voice market was falling after 2012 explained by the shift to data and messengers with cheaper package-pricing for data traffic and no interconnection and landing fees
  - General preference to interact in writing compared to making voice calls
- Mobile operators are seeking for new business models including the provision of more content and building of eco-systems, some still suggest taxing social network and messenger companies a traffic charge for consumer access to social media content

Global Mobile Voice Subscribers and Market Size Regression



Global Mobile Voice Market Size and Subscriber Base Development



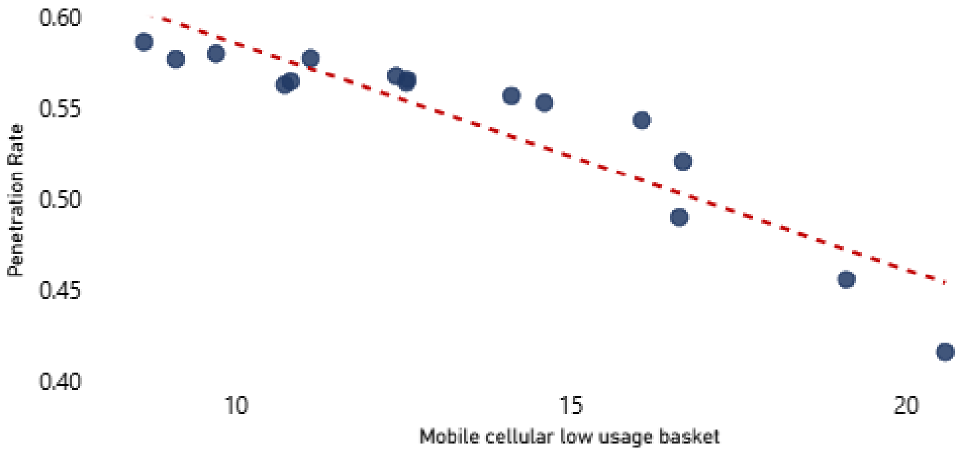
Note: correlation charts were produced manually in the preprocessing file because of their non-standard format

Source: ITU

Growth Drivers: ARPU

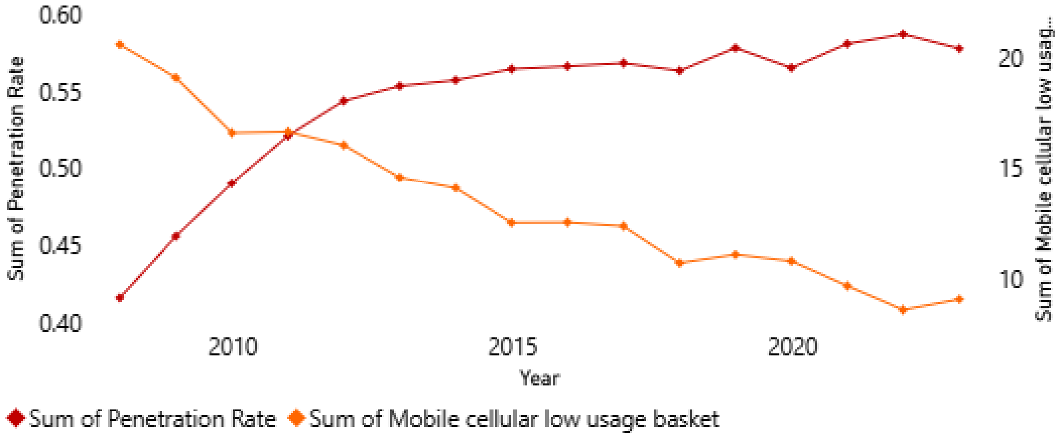
- Mobile voice ARPU decline over 2008-2023 was, unlike subscriber growth, the continuation of the trend
- ARPU was falling a the technology was becoming cheaper (Moore's law), economies of scale, transition to data traffic (VOLTE/VOIP) also in the emerging markets around 2008-2012, competition, and almost no government regulation, which is a strong advantage
- One should note that penetration rate growth well beyond 100% accounts for the users having several numbers, e.g., private and business and a share of business mobile numbers
- Correlation of ARPU decline and penetration rate growth was almost perfect - minus (0.95%)

Global Mobile Voice Penetration and ARPU



Note: correlation charts were produced manually in the preprocessing file because of their non-standard format

Global Mobile ARPU and Penetration Rate Development

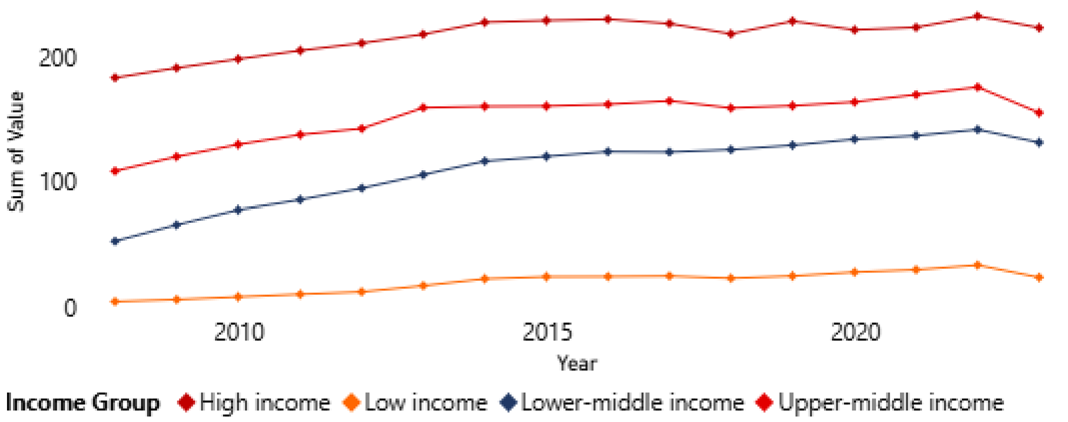


Source: ITU

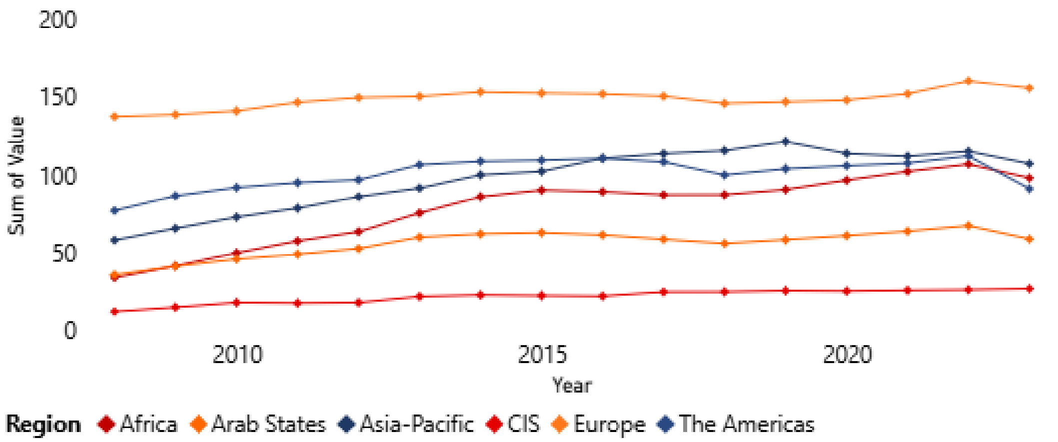
Mobile Voice Communications Penetration Rates Development

- Mobile voice penetration rates in higher income regions were above the world average drive by business segment and IOT, while this is still a less significant factor in lower income economies
- While the highest penetration rates in 2008-2023 in mobile voice market was in the CIS region the highest growth was in Asia and Africa
- High CIS penetration rates are explained by low ARPU, popularity of corporate phones and frequent change of tariffs and operators

Mobile Voice Penetration Rates in Key Economies by Income Split



Mobile Voice Penetration Rates in Key Economies by Regional Split



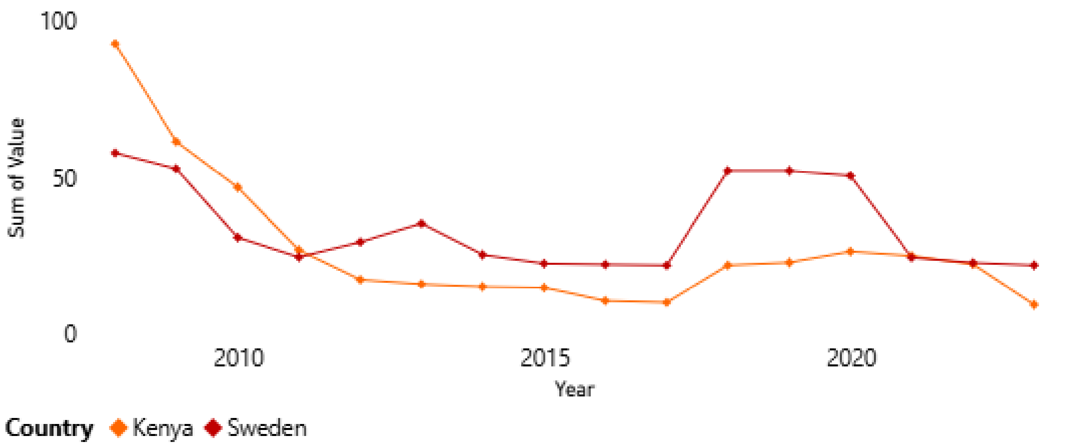
Source: ITU

Selected Developed vs. Emerging Comparison – Kenya : Sweden

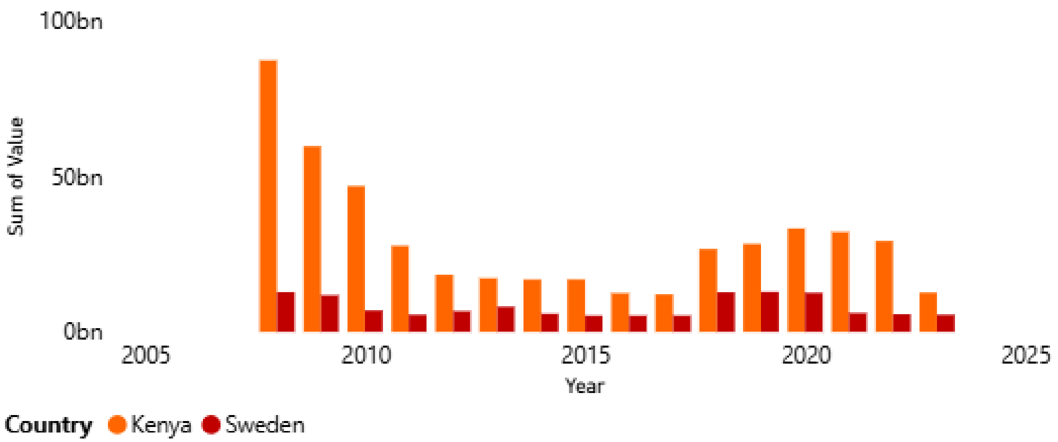
Except for high/low income divide and earlier market saturation Kenya and Sweden are consistent with the global mobile voice trends

- Despite a ~5.x difference in population the mobile voice market size in Kenya and Sweden is almost the same
- ARPU reflects a reducing gap between a developed and an emerging market
- Mobile voice market continued falling except around Covid-time in Kenya (there was no strict lockdown in Sweden)
- The shift to data started around 2008 or even earlier for both Kenya and Sweden
- Penetration rate in Kenya reached 100% only in 2019, Sweden's ARPU was stagnant over the period

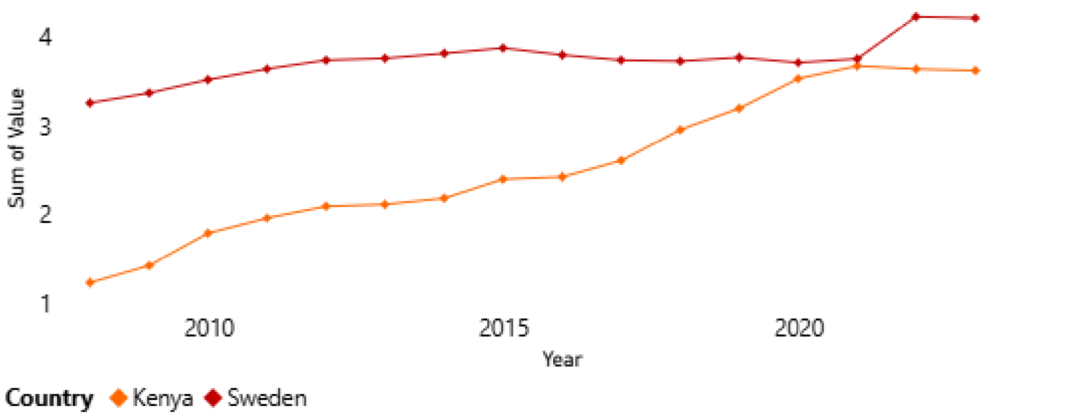
ARPU Development



Market Size Development



Penetration Rate



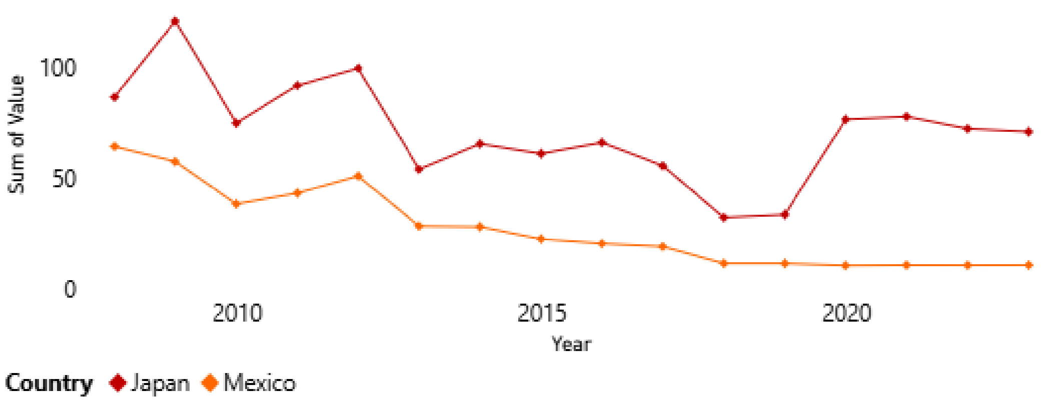
Source: ITU

Selected Developed vs. Emerging Comparison – Japan : Mexico

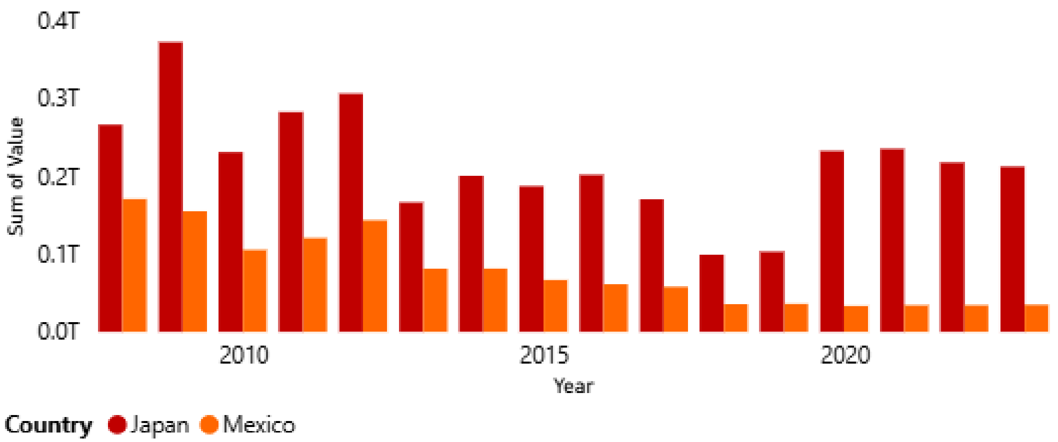
Mexico fits into the general mobile voice pattern, while Japan reflects stronger cultural preference for voice communication

- Japans market is larger than Mexico's by both population and market but the decline in mobile voice after 2012 is still visible (except Covid)
- ARPU development in Japan appears less consistent, the drop in 2008-2010 was probably due to subprime crisis and lower business activities but otherwise it is probably a cultural feature of Japan to have more voice communication
- Penetration rate growth in Japan is outstanding even for a developed market and may be an indicator of the future for the rest of the world
- Penetration rate in Japan reflect the culture of separate profile for business and private purposes, which seams both reasonable and ethical

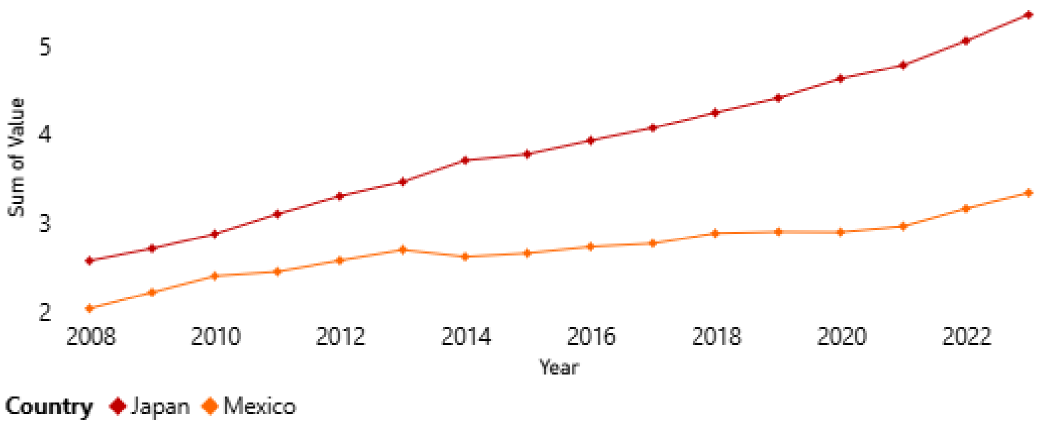
ARPU Development



Market Size Development



Penetration Rate



Source: ITU